

```
// Code-A1 (Fibonacci)
```

```
#include <iostream>
```

```
using namespace std;
```

```
int recursiveSteps = 0;
```

```
int iterativeSteps = 0;
```

```
// Recursive function for Fibonacci
```

```
int fibRecursive(int n) {
```

```
    recursiveSteps++;
```

```
    if (n <= 1)
```

```
        return n;
```

```
    return fibRecursive(n - 1) + fibRecursive(n - 2);
```

```
}
```

```
// Non-recursive (Iterative) Fibonacci
```

```
void fibIterative(int n) {
```

```
    if (n <= 0)
```

```
        return;
```

```
    int prev = 0, curr = 1, next;
```

```
    cout << prev << " ";
```

```
    if (n == 1)
```

```
        return;
```

```
    cout << curr << " ";
```

```
    for (int i = 2; i < n; i++) {
```

```
        next = prev + curr;
```

```
        cout << next << " ";
```

```
        prev = curr;
```

```
        curr = next;
```

```
        iterativeSteps++;
```

```
    }
```

```
}
```

```

int main() {
    int n;
    cout << "Enter the number of terms: ";
    cin >> n;
    cout << "\nFibonacci Series using Recursion: ";
    for (int i = 0; i < n; i++) {
        cout << fibRecursive(i) << " ";
    }
    cout << "\nTotal Recursive Steps: " << recursiveSteps;
    cout << "\n\nFibonacci Series using Iteration: ";
    fibIterative(n);
    cout << "\nTotal Iterative Steps: " << iterativeSteps;
    cout << endl;
    return 0;
}

```

// SAMPLE OUTOUT

```

/*
* Enter the number of terms: 5
*
* Fibonacci Series using Recursion: 0 1 1 2 3
* Total Recursive Steps: 19
*
* Fibonacci Series using Iteration: 0 1 1 2 3
* Total Iterative Steps: 3
*/

```